

Name: Prithvi S K K
Roll No: 2023115090
Mobile: 7448450523
Mail: prithviskk02@gmail.com

Appian Corporation is an American cloud computing and enterprise software company headquartered in McLean, Virginia, part of the Dulles Technology Corridor. The company sells a platform as a service (PaaS) for building enterprise software applications. It is focused on low-code development, process mining, business process management, and case management markets in North America, Europe, the Middle East, and Southeast Asia. Appian offers a low code platform with a visual interface and pre-built development modules. As with other low-code platforms, it enables businesses to create apps and automations using little or no code.

There are four rounds of interview

1. Coding round
2. Technical round
3. Techno - Behavioural round
4. HR round

Role: Software Engineer

Coding Round:

Candidates were sent a link to attend the online coding round, which was held on the Glider platform. We had to solve three questions in 75 mins.

Question 1: Given a sorted array and a value x , find the first and last occurrence of value x in the array and return them. If x is not found in the array, return $\{-1, -1\}$.

Link : <https://leetcode.com/problems/find-first-and-last-position-of-element-in-sorted-array/description/>

This is a well-known binary search problem.

Question 2: Given a sorted array which contains non – overlapping intervals and a new interval which need to be inserted into the appropriate position and return them . The answer should also need to be in sorted order, and there should not be any overlapping intervals .

Link : <https://leetcode.com/problems/insert-interval/description/>

Since the array is sorted, we can solve this question using a greedy approach

Question 3: Given a 1-D array which contains numbers from 1 to n where n is size of the array. It must contain a duplicate number and a missing number. A 2-D array in the size of $M * 2$ will be given, where M is the number of rows in the 2-D array and this 2-D array remains the same for every test case. Find the duplicate and missing number from the 1-D array and check whether both of them occur as a pair in the 2-D array.

If the pair occurs more than 1 time, return the pair with the string as “Multiple teams found”
Else if the pair occurs only 1 time, then return the pair with the string “A Team found”

Else return {-1 , -1 , “No team found”}

Link : <https://leetcode.com/problems/set-mismatch/description/>

This problem involves finding the duplicate and missing number .

I was able to solve all the problems with an optimal solution. Even if you are finding it difficult to find the optimal solution try brute force method, as submitting the test before time gives you an edge over others . 42 of us were shortlisted for the interview process.

Technical Round:

When I walked into the interview room, I'll admit I was a bit anxious at first. Thankfully, the interviewer's friendly demeanour helped me relax. We began with an introduction, after which he moved on to questions about the skills listed on my resume — and he went deep into the fundamentals.

Some of the questions I remember included:

- Distributed Systems: What is *consistency*? What do you think about the consistency in a distributed environment and type of consistency that MongoDB uses?
- Databases: Difference between SQL and NoSQL, and ACID vs BASE properties.
- Web Technologies: Why HTML replaced XML?
- JSON: How would an image be sent in JSON format? If it were sent as a string, how would you escape double quotes inside the string, and what is that process called?
- Networking: What's the average latency allowed between a client request and server response?
- HTTP: Difference between GET and POST; how would an HTTP request look?
- Security: What is JWT and how does it work internally? What are vulnerabilities?
- JavaScript: Difference between `setTimeout()` and `setInterval()`. How do you limit the number of times a `setInterval()` is executed?
- I was also asked to walk through the end-to-end flow of how a website works, right from the user's request to the server's response. including security aspects.

Then came the only DSA question:

"If there are crores of numbers in a stream where the range is [1,100], and only one number is not duplicated while the rest are duplicated multiple times, how would you find it optimally?"

I first suggested a frequency map (hash map) solution — $O(n)$ time, $O(n)$ space. He asked for further optimization. I proposed counting occurrences directly using an array/bitset, and since only one number is non-duplicated, we could maintain a counter — once we've seen duplicates for 99 numbers, the remaining one is the answer. This avoids extra space and works efficiently for the given range. He seemed satisfied with this approach.

Solution I suggested :

```
int count[101] = {0};
int duplicatesFound = 0;
for (number in stream) {
    count[number]++;
    if (count[number]==2){
        duplicatesFound++;
        if (duplicatesFound==99){
            for (int i = 1; i <= 100; i++) {
                if (count[i] == 1) {
                    return i;
                }
            }
        }
    }
}
```

This round lasted an hour and felt intense but rewarding.

Technical Round II:

This was more of a technical + behavioural conversation. This round was virtual .It started with my introduction and the interviewer inquired about my area of interest to which I answered “*Web Development*” and he asked some questions related to it , some of the question I could recall:

- Why is Next.js preferred over React?
- Difference between var, let, and const in JavaScript.
- Difference between Express.js and Node.js.
- Difference between TCP and UDP
- What are Web Sockets? What is the protocol that uses Web Sockets?

I was also asked “*Why Appian?*” to which I was able to answer confidently.

We also had some general web development discussions. The round lasted for about 15 minutes.

HR Round:

This round was casual and focused on my background, interests and some general ideas.The round lasted for about 10 minutes.

Finally the results were published around 9pm and I was fortunate to be one of them . 12 of us (8 from CEG and 4 from MIT were selected).

Bonus Tip:

At the end of each round, I was given the chance to ask my own questions. This is a great opportunity — prepare smart, interesting questions that can leave a lasting impression.

My Preparation and strategies:

- It's never too late to begin your DSA journey. Consistency matters more than speed — solving even one or two problems a day is enough if done regularly.
- Strengthen your core fundamentals in **OS, DBMS, and Computer Networks**. Platforms like **GeeksforGeeks** and **InterviewBit** are great resources to revise and practice these concepts.
- For structured DSA preparation, I recommend working through **Striver's A2Z Sheet**. If you enjoy reading in depth, explore "*Beyond Cracking the Coding Interview*" or curated content from **interviewing.io**.
- While solving problems, focus on recognizing **patterns** — most questions revolve around recurring approaches once you start identifying them.
- Be thorough with your **resume**. Every single word on it can lead to a question in the interview. Be transparent, and communicate your thoughts with confidence.